MATH I – Week 6

Fall 2026

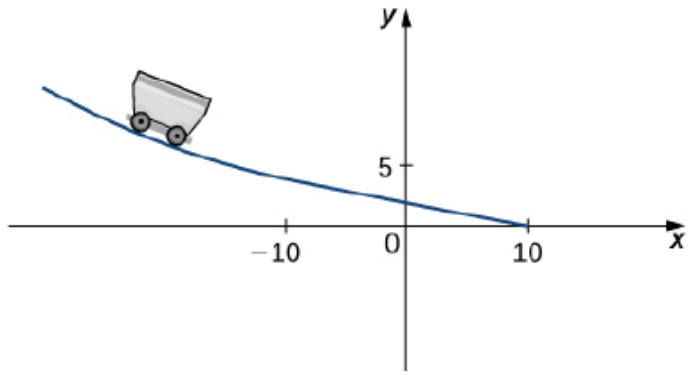
Practical Session exercises

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Exercise 1:

The function that describes the track is:

$$f(x) = \begin{cases} \frac{1}{10}x^2 + bx + c \text{ if } x < -10 \\ -\frac{1}{4}x + \frac{5}{2} \text{ if } x \geq -10 \end{cases}$$



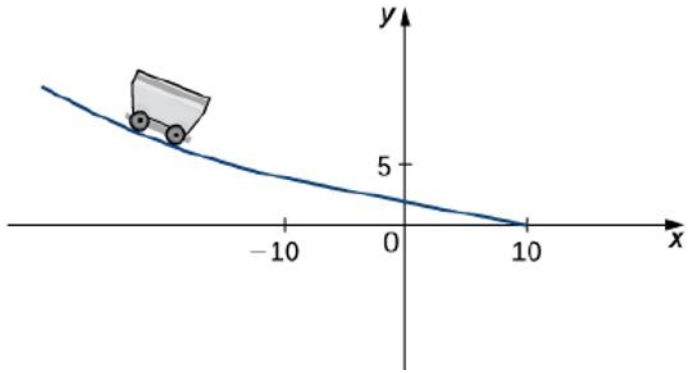
Is the motion of the car along the track smooth? Discuss (in terms of continuity and differentiability).

Exercise 2

EXAMPLE 3.14

A Piecewise Function that is Continuous and Differentiable

A toy company wants to design a track for a toy car that starts out along a parabolic curve and then converts to a straight line (Figure 3.17). The function that describes the track is to have the form $f(x) = \begin{cases} \frac{1}{10}x^2 + bx + c \text{ if } x < -10 \\ -\frac{1}{4}x + \frac{5}{2} \text{ if } x \geq -10 \end{cases}$ where x and $f(x)$ are in inches. For the car to move smoothly along the track, the function $f(x)$ must be both continuous and differentiable at -10. Find values of b and c that make $f(x)$ both continuous and differentiable.



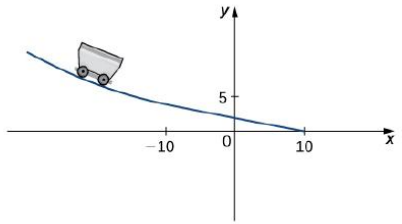


Figure 3.17 For the car to move smoothly along the track, the function must be both continuous and differentiable.

Solution

For the function to be continuous at $x = -10$, $\lim\limits_{x \to -10^-} f(x) = f(-10)$. Thus, since

$$\lim_{x \to -10^-} f(x) = \frac{1}{10}(-10)^2 - 10b + c = 10 - 10b + c$$

and $f(-10) = 5$, we must have $10 - 10b + c = 5$. Equivalently, we have $c = 10b - 5$.

For the function to be differentiable at -10,

$$f'(-10) = \lim_{x \to -10} \frac{f(x) - f(-10)}{x + 10}$$

must exist. Since $f(x)$ is defined using different rules on the right and the left, we must evaluate this limit from the right and the left and then set them equal to each other:

$$\begin{aligned}
\lim_{x \to -10^-} \frac{f(x)-f(-10)}{x+10} &= \lim_{x \to -10^-} \frac{\frac{1}{10}x^2+bx+c-5}{x+10} \\
&= \lim_{x \to -10^-} \frac{\frac{1}{10}x^2+bx+(10b-5)-5}{x+10} && \text{Substitute } c = 10b - 5. \\
&= \lim_{x \to -10^-} \frac{x^2-100+10bx+100b}{10(x+10)} && \text{Multiply numerator and denominator by 10.} \\
&= \lim_{x \to -10^-} \frac{(x+10)(x-10+10b)}{10(x+10)} && \text{Factor by grouping.} \\
&= b - 2.
\end{aligned}$$

We also have

$$\begin{aligned}
\lim_{x \to -10^+} \frac{f(x)-f(-10)}{x+10} &= \lim_{x \to -10^+} \frac{-\frac{1}{4}x+\frac{5}{2}-5}{x+10} \\
&= \lim_{x \to -10^+} \frac{-(x+10)}{4(x+10)} \\
&= -\frac{1}{4}.
\end{aligned}$$

This gives us $b - 2 = -\frac{1}{4}$. Thus $b = \frac{7}{4}$ and $c = 10\left(\frac{7}{4}\right) - 5 = \frac{25}{2}$.

Exercise 3

3. A reciprocal function

Let $f(x) = \frac{1}{x}, \qquad x \neq 0.$

a. Use the limit definition to find $f'(x)$.

b. Find the tangent line at $x = 2$.

c. Is the tangent slope positive or negative there? Interpret this geometrically.

Exercise 3 Solution

3. A reciprocal function

Let $f(x) = \frac{1}{x}, \qquad x \neq 0.$

a. Use the limit definition to find $f'(x)$.

b. Find the tangent line at $x = 2$.

c. Is the tangent slope positive or negative there? Interpret this geometrically.

For $x \neq 0$,

$$f'(x) = \lim_{h \to 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$$

$$= \lim_{h \to 0} \frac{-h}{hx(x+h)} = \boxed{-\frac{1}{x^2}}.$$

At $x = 2$, the point is $(2, \frac{1}{2})$, and the slope is $-\frac{1}{4}$. Thus,

$$y - \tfrac{1}{2} = -\tfrac{1}{4}(x - 2) \quad \Rightarrow \quad \boxed{y = 1 - \frac{x}{4}}.$$

The negative slope means that the graph falls as x increases near 2.

Exercise 4

Let $f(x) = |x|$.

a. Show that f is continuous at $x = 0$.

b. Use difference quotients to find $f'_-(0)$ and $f'_+(0)$.

c. Explain why $f'(0)$ does not exist.

d. Sketch the graph and identify the kink.

Exercise 4 Solution

Let $f(x) = |x|$.

a. Show that f is continuous at $x = 0$.
b. Use difference quotients to find $f'_-(0)$ and $f'_+(0)$.
c. Explain why $f'(0)$ does not exist.
d. Sketch the graph and identify the kink.

For $f(x) = |x|$,

$\lim_{x \to 0} |x| = 0 = f(0)$,

so f is continuous at 0.

The difference quotient is

$$\frac{|h|-0}{h} = \begin{cases} -1, & h < 0, \\ 1, & h > 0. \end{cases}$$

Hence,

$$\boxed{f'_-(0) = -1, \qquad f'_+(0) = 1}.$$

The one-sided derivatives differ, so $\boxed{f'(0) \text{ does not exist}}$. The V-shaped graph has a kink at the origin.

Exercise 5

Consider $f(x) = \begin{cases} x^2, & x \leq 1, \\ 2x - 1, & x > 1. \end{cases}$

a. Is f continuous at $x = 1$?

b. Calculate the left- and right-hand derivatives at $x = 1$.

c. Is f differentiable there?

d. Find the tangent line at the joining point, if it exists.

Exercise 5 Solution

Consider $f(x) = \begin{cases} x^2, & x \le 1, \\ 2x - 1, & x > 1. \end{cases}$

a. Is f continuous at $x = 1$?

b. Calculate the left- and right-hand derivatives at $x = 1$.

c. Is f differentiable there?

d. Find the tangent line at the joining point, if it exists.

At $x = 1$,

$$f(1) = 1, \qquad \lim_{x \to 1^-} f(x) = 1, \qquad \lim_{x \to 1^+} f(x) = 2(1) - 1 = 1.$$

Therefore, f is continuous there.

The one-sided derivatives are

$$f'_-(1) = 2(1) = 2, \qquad f'_+(1) = 2.$$

Thus, f is differentiable at 1, with

$$\boxed{f'(1) = 2}.$$

The tangent line through $(1, 1)$ is

$$\boxed{y = 2x - 1}.$$

Exercise 6
False or True? With justification

Determine whether each statement is true or false. Give a reason or counterexample.

a. If f is differentiable at a, then f is continuous at a.

b. If f is continuous at a, then f is differentiable at a.

c. If $f'(a) = 0$, then f must be constant.

d. If both one-sided derivatives exist and are finite, then $f'(a)$ exists.

e. A function can have a vertical tangent at a point where it is continuous.

f. A corner can occur where two smooth graph segments meet.

Exercise 6
False or True? With justification
Solution

Determine whether each statement is true or false. Give a reason or counterexample.

a. If f is differentiable at a, then f is continuous at a.
b. If f is continuous at a, then f is differentiable at a.
c. If $f'(a) = 0$, then f must be constant.
d. If both one-sided derivatives exist and are finite, then $f'(a)$ exists.
e. A function can have a vertical tangent at a point where it is continuous.
f. A corner can occur where two smooth graph segments meet.

Statement	Answer	Justification
a. Differentiability implies continuity	**True**	This is a standard theorem
b. Continuity implies differentiability	**False**	$f(x) = \|x\|$ at (0)
c. (f'(a)=0) implies (f) is constant	**False**	(f(x)=x^2) has $f'(0) = 0$
d. Finite one-sided derivatives imply a derivative exists	**False**	They must also be equal
e. A continuous function can have a vertical tangent	**True**	$f(x) = \sqrt[3]{x}$ at 0
f. Smooth segments can meet at a corner	**True**	Their slopes may differ at the joining point

Exercise 7

For each function, determine whether it is continuous and differentiable at 0. Identify any kink, cusp, or vertical tangent.

a. $f(x) = x^2$.
b. $g(x) = |x|$.
c. $p(x) = |x|^{2/3}$.
d. $q(x) = \sqrt[3]{x}$.

Exercise 7 Solution

For each function, determine whether it is continuous and differentiable at 0. Identify any kink, cusp, or vertical tangent.

a. $f(x) = x^2$.
b. $g(x) = |x|$.
c. $p(x) = |x|^{2/3}$.
d. $q(x) = \sqrt[3]{x}$.

Function	Continuous at 0?	Finite derivative at 0?	Behavior
x^2	Yes	Yes, $f'(0) = 0$	Horizontal tangent
$\|x\|$	Yes	No	Kink; slopes -1 and 1
$\|x\|^{2/3}$	Yes	No	Cusp; slopes approach $-\infty, +\infty$
$\sqrt[3]{x}$	Yes	No	Vertical tangent; slopes approach $+\infty$ on both sides

Exercise 8

Let $f(x) = x|x|$.

a. Write f as a piecewise function.

b. Use the derivative definition to find $f'(0)$.

c. Find $f'(x)$ for $x \neq 0$.

d. Is f' continuous at 0?

e. Does $f''(0)$ exist? Explain using one-sided limits.

Exercise 8 Solution

Let $f(x) = x|x|$.

a. Write f as a piecewise function.
b. Use the derivative definition to find $f'(0)$.
c. Find $f'(x)$ for $x \neq 0$.
d. Is f' continuous at 0?
e. Does $f''(0)$ exist? Explain using one-sided limits.

a. $f(x) = \begin{cases} -x^2, & x < 0, \\ x^2, & x \geq 0. \end{cases}$

b. At 0,

$$f'(0) = \lim_{h \to 0} \frac{h|h|}{h} = \lim_{h \to 0} |h| = \boxed{0}.$$

c.

$$f'(x) = \begin{cases} -2x, & x < 0, \\ 0, & x = 0, \\ 2x, & x > 0, \end{cases} = \boxed{2|x|}.$$

d. Since $2|x| \to 0 = f'(0)$, f' is continuous at 0.

e.

$$\frac{f'(h) - f'(0)}{h} = \frac{2|h|}{h}$$

has limits -2 from the left and 2 from the right. Therefore,

$$\boxed{f''(0) \text{ does not exist}}.$$

Exercise 9

Consider

$$f(x) = \frac{x^2-4}{x-2}, \qquad x \neq 2.$$

a. Simplify the expression while retaining its domain restriction.

b. Find $f'(x)$ on its domain.

c. Does the original function have a derivative at 2?

d. Define an extension at 2 that is differentiable there.

Exercise 9 Solution

Consider

$f(x) = \frac{x^2-4}{x-2}, \qquad x \neq 2.$

a. Simplify the expression while retaining its domain restriction.
b. Find $f'(x)$ on its domain.
c. Does the original function have a derivative at 2?
d. Define an extension at 2 that is differentiable there.

For $x \neq 2$,

$$f(x) = \frac{(x-2)(x+2)}{x-2} = \boxed{x+2}.$$

Therefore,

$$\boxed{f'(x) = 1, \qquad x \neq 2}.$$

The original function is undefined at 2, so it has no derivative there.

Define the extension by

$$\boxed{\widetilde{f}(2) = 4}.$$

Then $\widetilde{f}(x) = x + 2$ for all real x, and $\widetilde{f}'(2) = 1$.

Correcting common errors

Exercise 10

A student writes:

a. $\frac{d}{dx}(x^2x^3) = (2x)(3x^2) = 6x^3.$

b. $\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \frac{2x}{1}.$

c. $\frac{d}{dx}(7x^3 + 5) = 21x^2 + 5.$

For each statement:

- Identify the error.
- Give the correct derivative.
- State the relevant differentiation rule.

Correcting common errors *Exercise 10 Solution*

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b. $\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \frac{2x}{1}.$

c. $\frac{d}{dx}(7x^3 + 5) = 21x^2 + 5.$

For each statement:

- Identify the error.
- Give the correct derivative.
- State the relevant differentiation rule.

a. The derivative of a product is not the product of the derivatives:

$$\frac{d}{dx}(x^2x^3) = (2x)x^3 + x^2(3x^2) = \boxed{5x^4}.$$

b. The derivative of a quotient is not the quotient of the derivatives:

$$\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \boxed{\frac{x^2 + 2x - 1}{(x+1)^2}}, \qquad x \neq -1.$$

c. The derivative of a constant is zero:

$$\frac{d}{dx}(7x^3 + 5) = \boxed{21x^2}.$$

Exercise 11

Let

$$f(x) = \frac{1}{x}, \qquad x \neq 0.$$

a. Find $f'(x)$, $f''(x)$, and $f'''(x)$.

b. Identify the pattern in signs and coefficients.

c. Conjecture a formula for $f^{(n)}(x)$.

Exercise 11 Solution

Let

$$f(x) = \frac{1}{x}, \qquad x \neq 0.$$

a. Find $f'(x)$, $f''(x)$, and $f'''(x)$.

b. Identify the pattern in signs and coefficients.

c. Conjecture a formula for $f^{(n)}(x)$.

$$\boxed{f'(x) = -\frac{1}{x^2}, \qquad f''(x) = \frac{2}{x^3}, \qquad f'''(x) = -\frac{6}{x^4}.}$$

The signs alternate, and the coefficient magnitudes are factorials. The general formula is

$$\boxed{f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}}, \qquad x \neq 0.$$

Exercise 12

During a heating experiment,

$$T(t) = 20 + 8t - t^2, \qquad 0 \le t \le 4,$$

where T is measured in degrees Celsius and t in minutes.

a. Find $T'(t)$.
b. Interpret $T'(1)$ and $T'(3)$.
c. Find the average heating rate over $[1, 3]$.
d. Use $T(2)$ and $T'(2)$ to estimate $T(2.1)$.
e. Calculate the exact $T(2.1)$ and the estimation error.

Exercise 12 Solution

During a heating experiment,

$$T(t) = 20 + 8t - t^2, \qquad 0 \le t \le 4,$$

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d. Use $T(2)$ and $T'(2)$ to estimate $T(2.1)$.
e. Calculate the exact $T(2.1)$ and the estimation error.

a. $\boxed{T'(t) = 8 - 2t}$.

b. $\boxed{T'(1) = 6^\circ\text{C/min}, \qquad T'(3) = 2^\circ\text{C/min}}$.

The component is heating at both times, but more slowly at $t = 3$.

c. $T(1) = 27, \qquad T(3) = 35,$

so

$$\boxed{\frac{35 - 27}{3 - 1} = 4^\circ\text{C/min}}.$$

d. Since $T(2) = 32$ and $T'(2) = 4$,

$$T(2.1) \approx 32 + 4(0.1) = \boxed{32.4^\circ\text{C}}.$$

e. $T(2.1) = 20 + 8(2.1) - (2.1)^2 = \boxed{32.39^\circ\text{C}}$.

The tangent estimate is too high by

$$\boxed{0.01^\circ\text{C}}.$$